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# Application of deep learning in workpiece defect detection

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**Abstract**

Due to problems in manufacturing process and forging technology, some of the workpieces production in the workshop have defects, which affects the ornamental value and utility of the workpieces. At present, the detection of workpiece defects is still mainly relying on manual detection, which consumes human resources and has a high detection error rate. To solve the above problems, deep learning is used to improve the detection of workpiece defects and reduce the cost of workpiece production. This article uses Faster R-CNN[1] as the basic architecture and integrates Resnet50[2] as the backbone network. In addition, Deformable Convolutional Networks[3] is introduced to extract defect features of different forms, and the multi-scale transformation problem is solved through Feature Pyramid Networks[4]. Comparing with YOLOv3[5], Faster R-CNN, Mask R-CNN[6] and other models, the results show that the mAP of the object detection model based on Faster R-CNN is 89.21%, the detection effect is the best, and it can meet the actual production needs.



*Keywords: Defect detection; Faster R-CNN; deformable convolution; feature pyramid;*

## 1. Introduction

With the continuous development of modern manufacturing, rapid prototyping technology is becoming more mature and widely used in industrial production. Due to the existence of casting process, production process, material quality and other problems, some of the produced parts have defects. The market value of artworks, souvenirs and other ornamental products whose main evaluation criterion is appearance quality. If there is a defect in the appearance, it is a waste product. For practical products such as steel and hardware products whose main

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evaluation criteria are use value, their market value is determined by use value. The workpiece's defects will directly affect the safety and stability of later use, the workpiece is scrap.

In the current actual production, the objection of workpiece defects is still mainly by manual inspection. For ordinary products, workers on the production line pick out defective products through visual observation. When there are a large number of waste products, certain interventions and adjustments are made to the production process to eliminate production equipment failures and reduce the rejection rate. For products with a high pass rate, professional quality inspection staff use professional instruments to carefully measure and inspect the produced products to improve product quality. Therefore, the labor's manual inspection greatly increases the cost of workpiece production. In addition, due to a certain error rate in manual detection, failures can not be found in time and losses can be reduced. Therefore, it is of great practical significance to improve the detection method of workpiece defects, increase production efficiency and control quality accuracy.

This research aims to improve the Faster R-CNN deep neural network, so as to automatically detect whether the workpiece has defects, improve the accuracy of workpiece defect detection, and reduce the cost of workpiece production.

## 2. Defect detection model based on Faster R-CNN

This article mainly uses deep learning methods for defect detection. For the small defect area, the close background color of the defect, and the large difference in the shape of the same type of defect, as shown in Figure 1 (a) below, ordinary neural networks cannot extract defect features well[7]. To solve the above problems, Faster R-CNN[1] is used as the basic architecture, and Resnet50[2] is integrated as the backbone network. In addition, Deformable Convolutional Networks[3] is introduced to extract defect features with different shapes, and the Faster R-CNN multi-scale problem is solved through Feature Pyramid Networks[4] (FPN). The network structure diagram of the workpiece defect detection model is shown in Figure 1 (b) below.

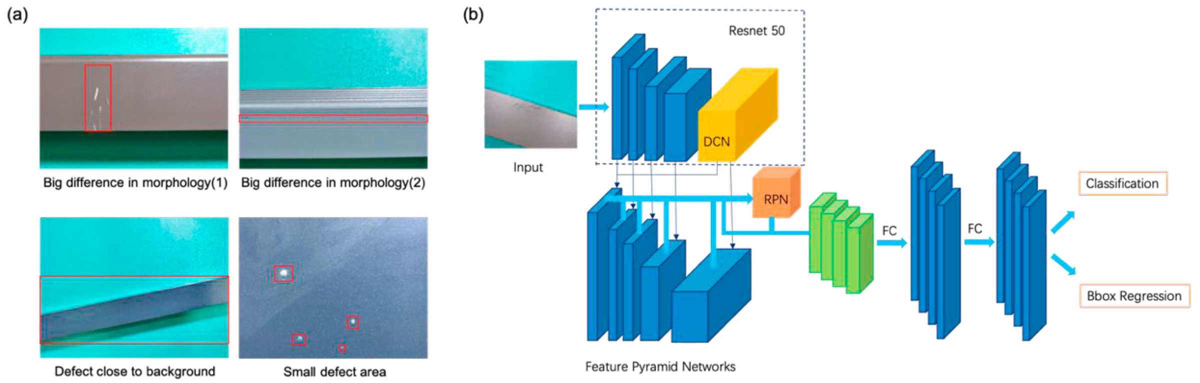


Fig. 1. (a) Schematic diagram of workpiece defect; (b) Network structure diagram of workpiece defect detection model.

### 2.1. Introduction to Faster R-CNN

In 2016, Ross Girshick and others proposed Faster R-CNN[1], which generates Region Proposal through a regional recommendation network, which solves the problem of the slow speed of the Selective Search method in R-CNN and Fast R-CNN to find Region Proposal. First, the feature map is obtained by pre-training convolutional neural network, and the extracted feature map is processed by the regional recommendation network to obtain the Region Proposal of each candidate frame. Finally, the SVM discriminator is used to determine the category of the candidate frame, and the Bbox regressor is used to further adjust its position.

### 2.2. Backbone network

In the defect detection model, the convolutional neural network is used as a feature extraction network, which has a vital impact on the later prediction results. Generally speaking, as the number of network layers increases, the effect of feature extraction is better, but when the network has reached the optimal state, as the network deepens, the accuracy of the model decreases. In 2015, Dr. Kaiming He proposed a deep residual network to solve the above problems. In this paper, considering the accuracy of the model and the operating speed, ResNet50[2] is selected as the backbone network. Due to the uncertain shape of the defect, the traditional regular square structure convolution is not strong enough to handle defects of this shape. On the basis of the ResNet50 network, the Deformable Convolution Network[3] (DCN) is introduced.

Traditional convolution is based on a defined filter size and operates on a predefined rectangular grid of the input image. In DCN, the grid is deformable, and the displacement variable of each point can be automatically calculated during the convolution calculation process. In this way, features are taken at the most suitable place for convolution. It is equivalent to the scalable change of each square of the convolution kernel, which changes the range of the receptive field, and the receptive field becomes a polygon. The specific calculation is shown in formula 1.

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \tag{1}$$

Where $p_0$ represents each position in the output feature map y; $p_n$ is an enumeration of grid R positions, $\Delta p_n$ and represents the offset $\Delta p_n = n = 1,\ldots,N$ , where $N = |R|$ .It is necessary to sample on irregular and offset $p_n + \Delta p_n$ , and the offset $\Delta p_n$ is usually a fraction, which is realized by bilinear interpolation, as shown in Equation 2.

$$x(p) = G(q, p) \cdot x(q) \tag{2}$$

Among them, $p = p_0 + p_n + \Delta p_n$ , q is selected from all positions of the feature map x, and $G(q, p)$ is the kernel function of bilinear interpolation. Since the deformable convolution is performed at the 2D latitude of the feature map, it can be divided into two kernel functions, such as the formula 3 shown.

$$G(q, p) = g(q_x, p_x) \cdot g(q_y, p_y) \tag{3}$$

Among them, $g(a, b) = \max(0, 1 - |a - b|)$.

As shown in Figure 2 (a) below, an additional convolutional layer is used to learn the offset and share the input feature map. Convolution kernels for offset and output features are trained at the same time, and the gradient can be propagated backwards through bilinear interpolation. Since DCN needs to learn the offset based on the previous convolution feature map, the last block of the original ResNet50 can be changed to variable convolution.

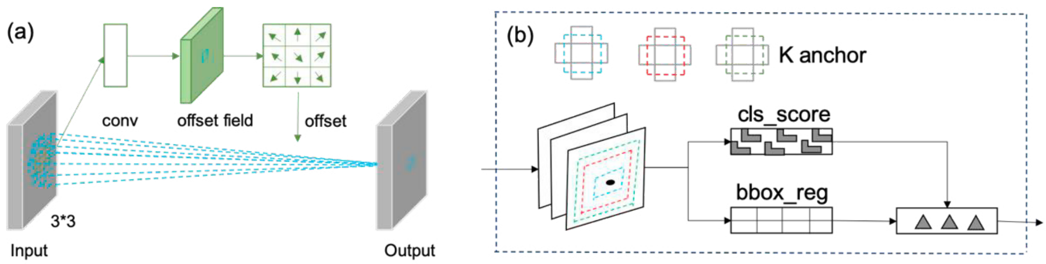


Fig. 2. (a) Deformable convolution diagram; (b) RPN network structure diagram.

## 2.3. Region Proposal Network

Before Faster R-CNN, Selective Search was commonly used to extract candidate frames, which took a long time to extract an image. Kaiming He and others proposed to extract candidate frames on the feature map by adding Region Proposal Network[1] (RPN).

The RPN network uses a sliding window mechanism to extract candidate frames on the feature map. Each sliding window position generates K candidate windows of different scales, and extracts corresponding anchor features for target classification and frame regression. The RPN network structure is shown in Figure 2 (b) above.

In training, RPN network uses multi-task loss, and its loss function is the sum of classification loss and regression loss. The fully connected layer used to determine the category of the candidate frame calculates the k+1 dimensional array P through softmax, which represents the probability of belonging to the k category and the background, as shown in formula 4.

$$\mathrm{P} = \left( p_0, p_1, \dots p_k \right) \tag{4}$$

The fully connected layer used to adjust the position of the candidate frame outputs a 4*k-dimensional array t, which represents the translation and scaling parameters of the candidate frame when they belong to class k category.

$$t_k = \left( t_{kx}, t_{ky}, t_{kw}, t_{kh} \right) \tag{5}$$

Where k represents the index of the category to which the candidate frame belongs; represents the translation information of the candidate frame on the X axis and the Y axis; represents the zoom information of the length and width of the candidate frame. The overall loss function of RPN is shown in Equation 6.

$$\mathrm{L}\left( \{p_i\}, \{t_i\} \right) = \frac{1}{N_{cls}} \sum_i L_{cls}\left( p_i, p_i^* \right) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* L_{res}\left( t_i, t_i^* \right) \tag{6}$$

Among them, $p_i$ represents the predicted probability of the target that anchor frame i belongs to; $p_i^*$ is 1 represents the target i to which anchor frame i belongs, and $p_i^*$ represents 0 that the anchor frame i does not belong to target i; $t_i$ represents the coordinates of the predicted candidate frame; $t_i^*$ represents the true candidate frame The target; $\lambda$ represents the balance weight parameter; $L_{cls}\left( p_i, p_i^* \right)$ represents the classification loss; $L_{res}\left( t_i, t_i^* \right)$ represents the regression loss.

$$L_{cls}\left( p_i, p_i^* \right) = -log\left[ p_i^* p_i + \left( 1 - p_i^* \right)\left( 1 - p \right) \right] \tag{7}$$

$$L_{res}\left( t_i, t_i^* \right) = \sum_{j=1}^{4} smooth_{L_1}\left( t_{ij} - t_{ij}^* \right) \tag{8}$$

$$smooth_{L1}(X) = \begin{cases} 0.5 \cdot \mathrm{x}^2 & if\ |x| < 1 \\ |x| - 0.5 & otherwise \end{cases} \tag{9}$$

## 2.4. Feature pyramid networks

After extracting features through the backbone network Resnet50, the space size is reduced by 16 times compared with the original picture. The features proposed by the candidate frame that are too small are not discriminative, which affects the performance of the model to a certain extent. To solve this problem, Feature Pyramid Networks[4] (FPN) is introduced to improve the network. Since the low-level features contain structural information, and the high-level features contain semantic information, the low-level features are convolved and upsampled and then merged with the high-level information to enhance feature expression. The specific structure is shown in Figure 3 (a) above.

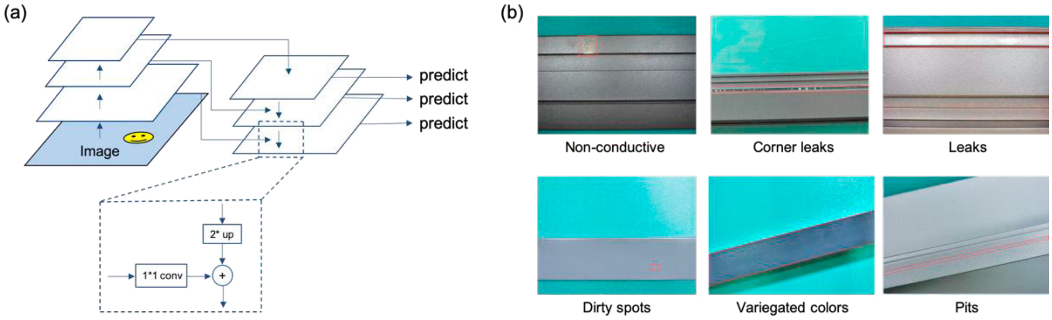


Fig. 3. (a) Feature pyramid network structure diagram; (b) Picture of surface defects of aluminum profile.

## 3. Experimental results and analysis

### *3.1. Experimental data*

The experiment uses part of the data from the aluminum profile surface defect recognition contest[8] as a data set, and selects 6 types of defects, which are dirty spots, variegated colors, pits, leaks, corner leaks, and non-conductivity. Figure 4 (b) shows different types of defect pictures and corresponding defect location marking boxes. In addition, there is a large sample imbalance between different types of defects. In order to solve this problem, this paper uses flipping, rotation and other methods to amplify the data to balance the defective samples. Then the data set is divided into three parts according to the ratio of 6:2:2, which are training data, verification data and test data.

### *3.2. Evaluation index*

This article uses AP and mAP commonly used in target detection models as evaluation indicators to measure the pros and cons of the model[9]. The formula is as follows

$$\mathrm{AP} = \int_0^1 p(r)\, d_r \tag{10}$$

Where p represents the accuracy rate, r represents the recall rate; calculating AP is to calculate the area under the P-R curve.

$$\mathrm{mAP} = \frac{1}{N} \sum AP_i \tag{11}$$

Where N represents the number of categories; mAP is the average value of AP values in all categories.

### *3.3. Model comparison*

The hardware and software configuration of the laboratory is shown in Table 1, using Python programming language and PaddlePaddle framework for experiments.

Table 1. Hardware environment and software environment.

| Hardware environment | Software environment |
|---|---|
| Processor : Xeon(R) Gold 6271C | Operating system : Linux-4.4.0 |
| ARM :32G; Disk：100GB | Programming language : Python3.7 |
| GPU : Tesla V100 16G | Frame : PaddlePaddle 1.8.4 |

In the same data set and experimental environment, using the YOLOv3, Faster R-CNN, Mask R-CNN and the improved Faster R-CNN model, and compare the performance. The comparison data of mAP and detection time of different models are shown in Table 2 above.

Table 2. Comparison of different detection algorithms.

| | mAP/% | Detection time /s |
|---|---|---|
| YOLOv3 | 75.82 | 0.461 |
| Faster R-CNN | 79.13 | 0.551 |
| Mask R-CNN | 81.25 | 0.768 |
| Improve Faster R-CNN | 89.21 | 0.573 |

It can be seen from the table that the detection speed of YOLOv3 is faster, but the mAP is lower, and the mAP of Mask R-CNN is higher but the detection speed is slower. Compared with YOLOv3 and Mask R-CNN, the mAP and detection speed of traditional Faster R-CNN are moderate. The improved model mAP in this article is higher than Mask R-CNN and the detection speed is faster, fully experiencing the advantages of the model.

As shown in Figure 4 below, the detection effects of different defects are different, and their AP values are also different. The AP values of corner leaks, leaks, pits and variegated are higher, all at 94% and above, while the AP of dirty spots The value is low, only 54.83%. It can be seen that the model has a low detection accuracy for defects in a small area, and there is a risk of misjudgment.

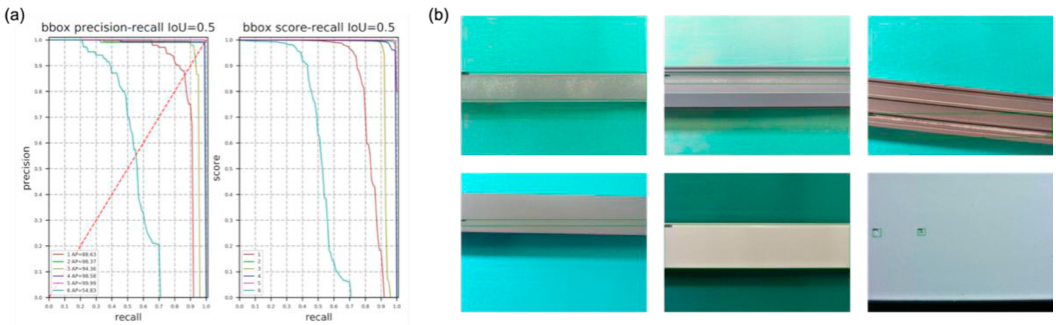


Fig. 4. (a) PR curves of different defects; (b) Defect detection model prediction effect map.

## 4. Summary and outlook

Based on the Faster R-CNN model, this paper uses ResNet50 and deformable convolution to extract pre-processed image features, combines feature pyramids to solve the problem of scale changes, and builds a workpiece defect detection model. And use the aluminum profile surface defect recognition contest as a data set for model training to extract workpiece defects, and the mAP of the model is 89.21%. From the AP value of each category, the detection of small defects needs to be improved, and the detection of small defects should be further studied in the later stage to improve the accuracy of its detection.

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